

MATĚJ TVARŮŽKA

Prague, Czech Republic | matej.tvar@gmail.com | +420 777 953 501

EDUCATION

Charles University, Faculty of Science, Prague, CZ

Master's Degree in Ecology

2025 – Present

- Thesis: "Secondary syntopy in North American passerines"
- Supervisor: Prof. Vladimír Remeš
- Focus: Evolutionary drivers of species coexistence at multiple spatial scales

Charles University, Faculty of Science, Prague, CZ

Bachelor's Degree in Ecology and Evolutionary Biology

2022 – 2025

- Thesis: "Ecology and evolution of secondary sympatry in birds"
- Supervisor: Prof. Vladimír Remeš

RESEARCH & FIELD EXPERIENCE

Woodpecker Hybrid Survey, Czech Republic, Prague, CZ

Field Research Assistant (Volunteer)

April 2025

- Conducted field mapping of hybrids between Syrian and Great Spotted Woodpeckers under the direction of Prof. Łukasz Kajtoch.

Independent Field Research Project (SOČ), Opava, CZ

Lead Student Researcher

2020 – 2022

- Awarded 7th place in the category "Nature Conservation" in the National Round of the Czech Republic Science Activity Competition.
- Designed and executed a 2-year breeding season mapping project focusing on urban woodpecker assemblages in Opava.

TECHNICAL & QUANTITATIVE SKILLS

- Statistical Modeling: Advanced proficiency in GLM, GAM, and Bayesian inference (experience utilizing 'Statistical Rethinking' framework and JAGS).
- Spatial Data Science: Advanced spatial data analysis in Python (geopandas, seaborn, rioxarray, xvec, libpysal) and R (sf, vegan, tidyverse). Experienced with QGIS workflows.
- Relevant Advanced Coursework: Macroecology, Spatial Data Science for Social Geography, Evolutionary Genetics, Evolutionary Analysis, Tropical Ecology.
- Languages: Czech (Native), English (Proficient / B2-C1 level academic reading and writing), German (A2).

UPCOMING PROFESSIONAL DEVELOPMENT

Math-Bio Exchange Summer School

September 2026

- Course on modelling of biological processes, hosted by the University of Warsaw.

Science School in Quantitative Ecology

September 2026

- Advanced upcoming coursework at the University of Bayreuth.